

Ligjerata 6e

Energjia kinetike e rrotullimit

$$E_{krot} = \frac{m v^2}{2} = \frac{m (r \omega)^2}{2} = \frac{m r^2 \omega^2}{2} = \frac{I \omega^2}{2}$$

$I = m r^2$

$v = r \cdot \omega$

$$E_{krot} = \frac{1}{2} m_1 r_1^2 \omega^2 + \frac{1}{2} m_2 r_2^2 \omega^2 + \dots + \frac{1}{2} m_n r_n^2 \omega^2$$

$$E_{krot} = \frac{1}{2} (m_1 r_1^2 + m_2 r_2^2 + \dots + m_n r_n^2) \omega^2$$

$$I_1 = m_1 r_1^2$$

$$I_2 = m_2 r_2^2$$

$$I_n = m_n r_n^2$$

$$E_{krot} = \frac{1}{2} (I_1 + I_2 + \dots + I_n) \omega^2$$

$$I = I_1 + I_2 + \dots + I_n$$

$$E_{krot} = \frac{1}{2} I \cdot \omega^2$$

